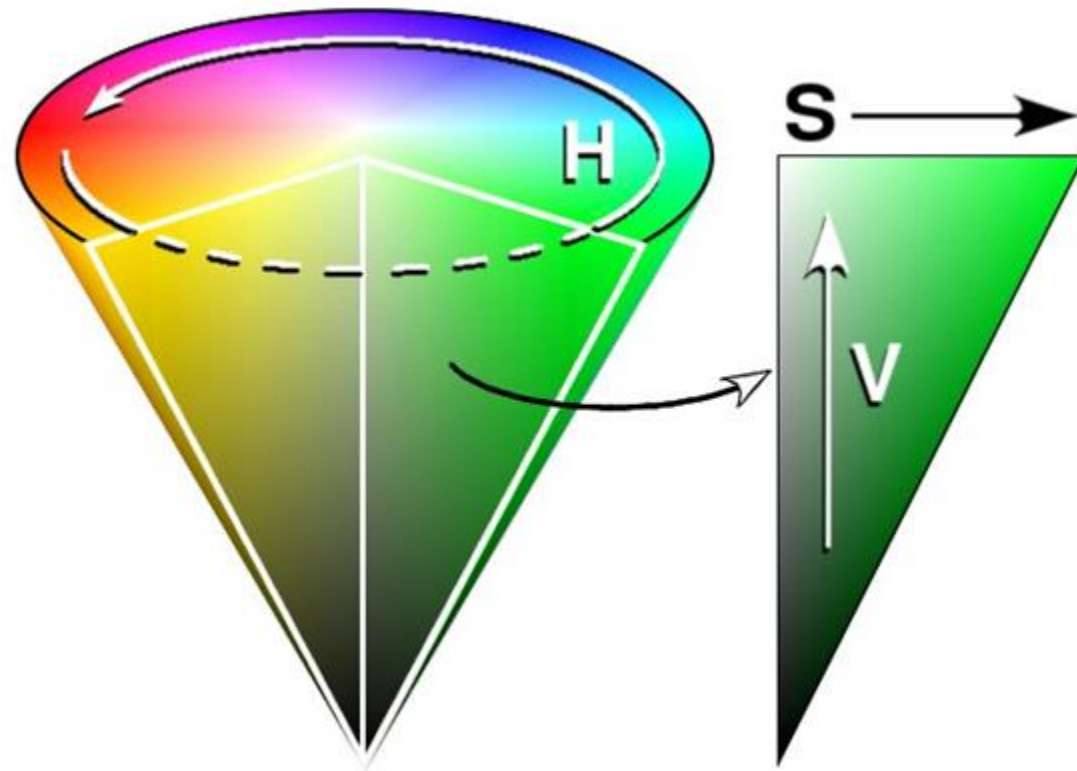


Image Processing - Lecture 9



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Reading:

Szeliski - Sec 2.3.2 + 3.1.2 (basic only)

Gonzalez / Woods - Sec 6.1 - 6.4

Solomon / Breckon - Sec 1.4 + 3.2.3 + 3.4.7

Colour Spaces & Processing Colour Images

Recap: Colour Images



What value(s) will I get at **this** pixel?

Recap: Colour Images



R



B



G

Do my colour channels look like this?

Recap: Colour Images



R



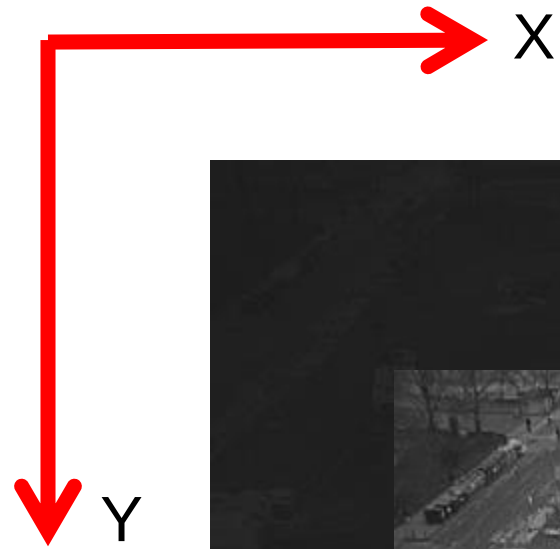
B



G

My colour channels look like this!

Recap: Colour Channels



■ RGB colour representation

- Red, Green and Blue light
- each (x, y) pixel = $\{r, g, b\}$ integer vector
- colour image = 3-channel image
 - (x, y, c) for $c = \{0, 1, 2\}$

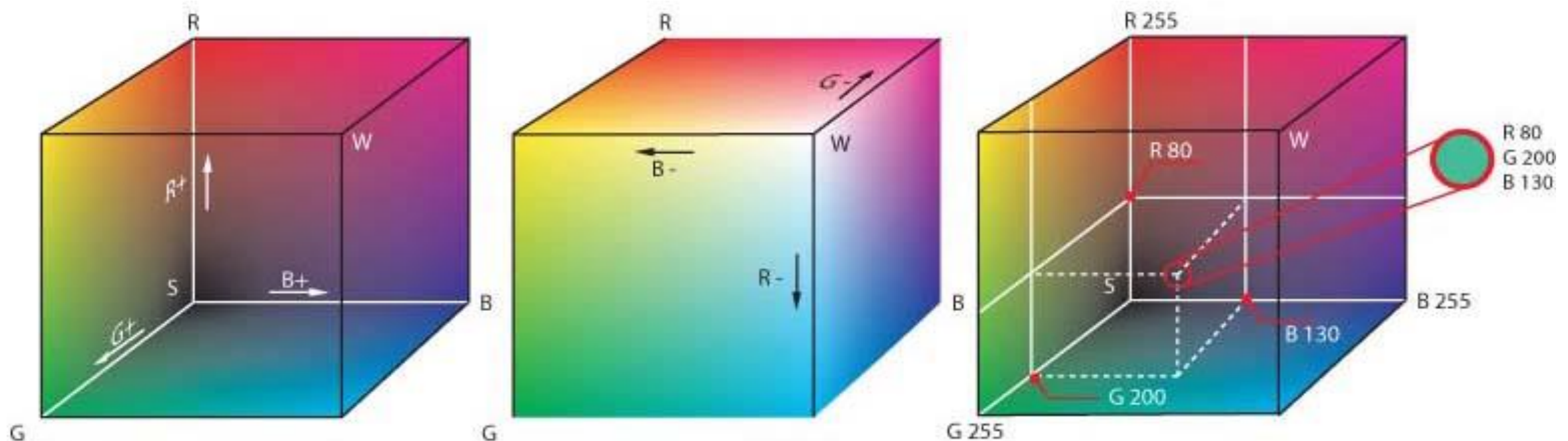
colour
channels (r,g,b)

**Each channel has 1 value, hence
each channel is a greyscale image
(combined to form colour).**

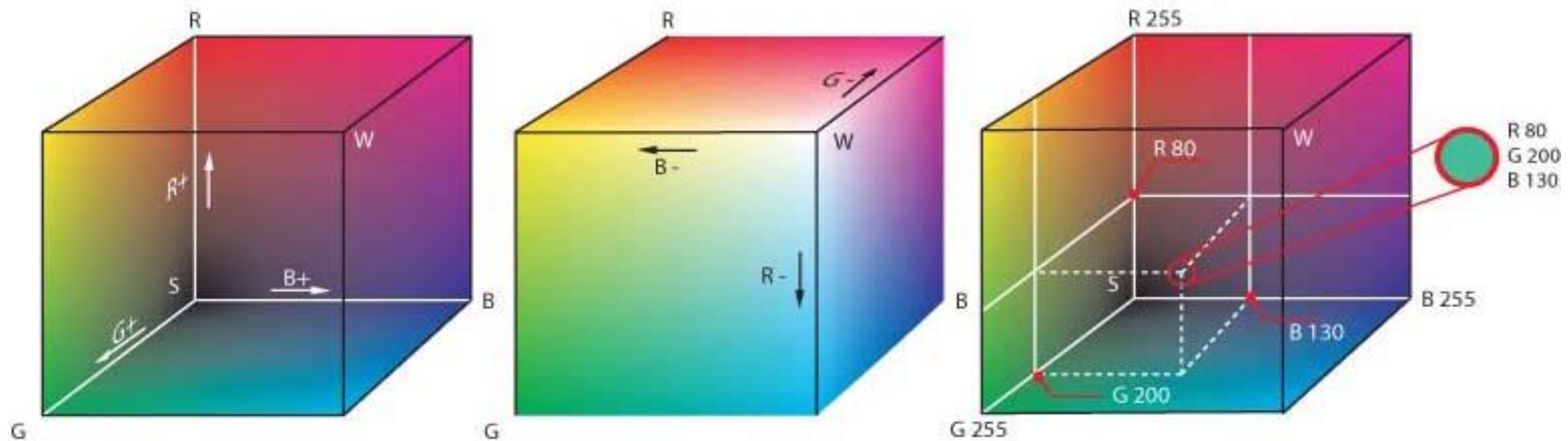
Image processing in and on colour !

RGB Colour Space

- Colours represented as R, G, B intensities.
 - The space is modelled geometrically by a cube with axes R, G and B.
 - If each edge is $0 \rightarrow 255$ (8-bit colour)
 - black = $(0,0,0)$ (origin) - white = $(255,255,255)$ (opposite corner)



RGB Colour Space

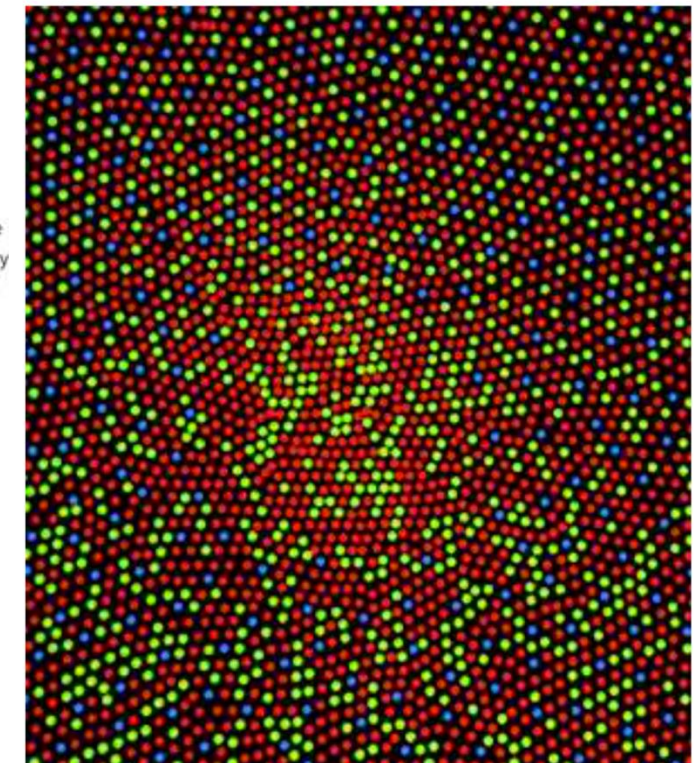
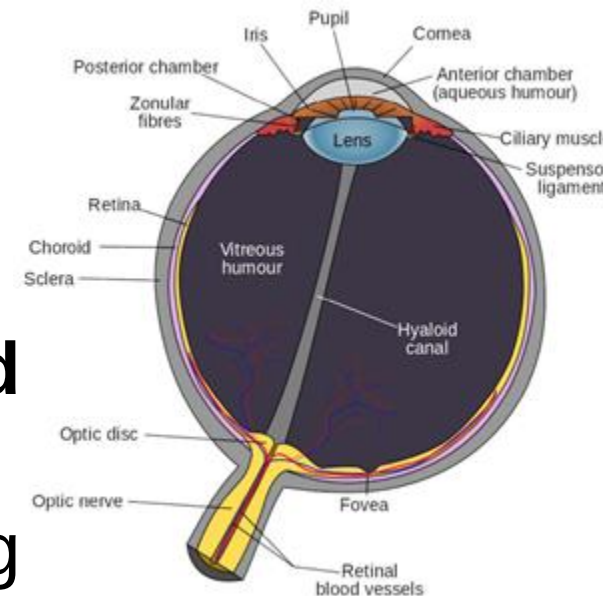


RGB is perceptually non-linear.

- Starting at **white** and subtracting the **blue** component produces... **yellow!**
 - Starting at **white** and subtracting the **red** component produces... **blue!**
 - Starting at **red** and adding the **blue** component produces... **pink!**
- ❑ RGB representation does not follow a natural representation of colours (as humans would reason about them)

Primer: Human Eye

- Photoreceptors:
 - Along the retina - stamp-sized piece of tissue at the back of the eye
 - **rods** and **cones**, scattered non-uniformly.
 - ~120 million *rods* - **low light intensity**
 - ~6 million *cones* - **colour sensitivity**
 - central fovea only has cones
- Types of photoreceptors and their distribution **influence brightness and colour perception**
 - e.g., periphery is good at detecting motion but not colour



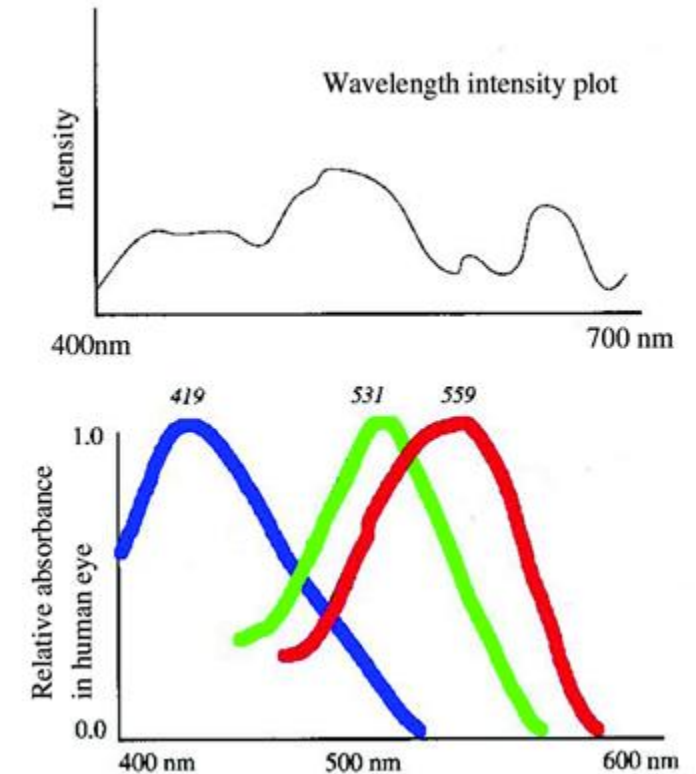
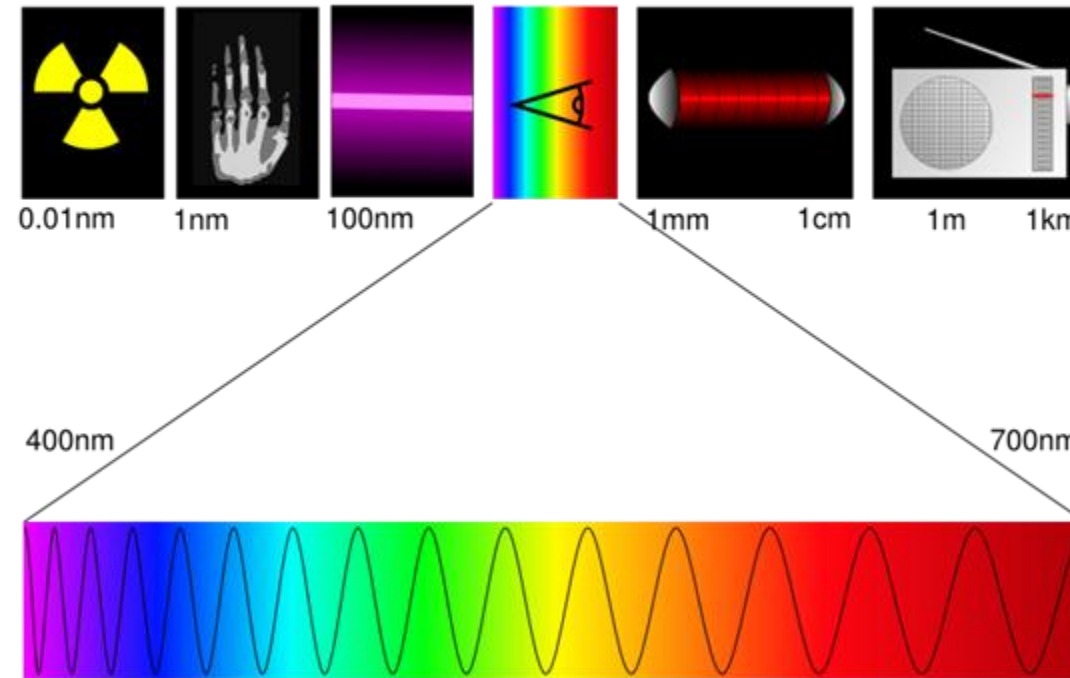
The human fovea

Why RGB?

- Based on human perception of visible spectrum.

- Human eye

- Long, Medium and Short* wavelength receptors corresponding roughly to **Red**, **Green** and **Blue**.
- We see colour as variable combinations of these three *primary* colours.



Right: how a dog sees the world!

Why Not RGB?



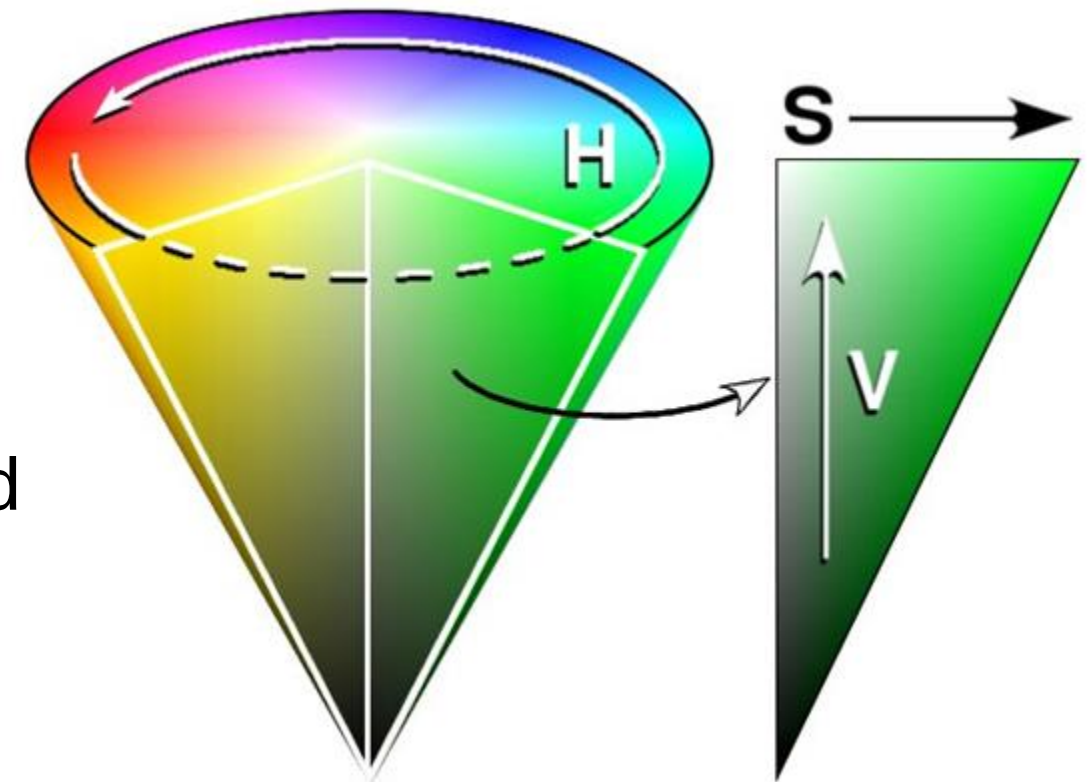
Individual RGB colour channels – **not** great for colour separation of image regions.

HSV Colour Space

Colours represented in their **H, S, V parametrised space**

- Commonly modelled as a cone.
- A more “**perceptual colour space**” – *better follows human intuition.*

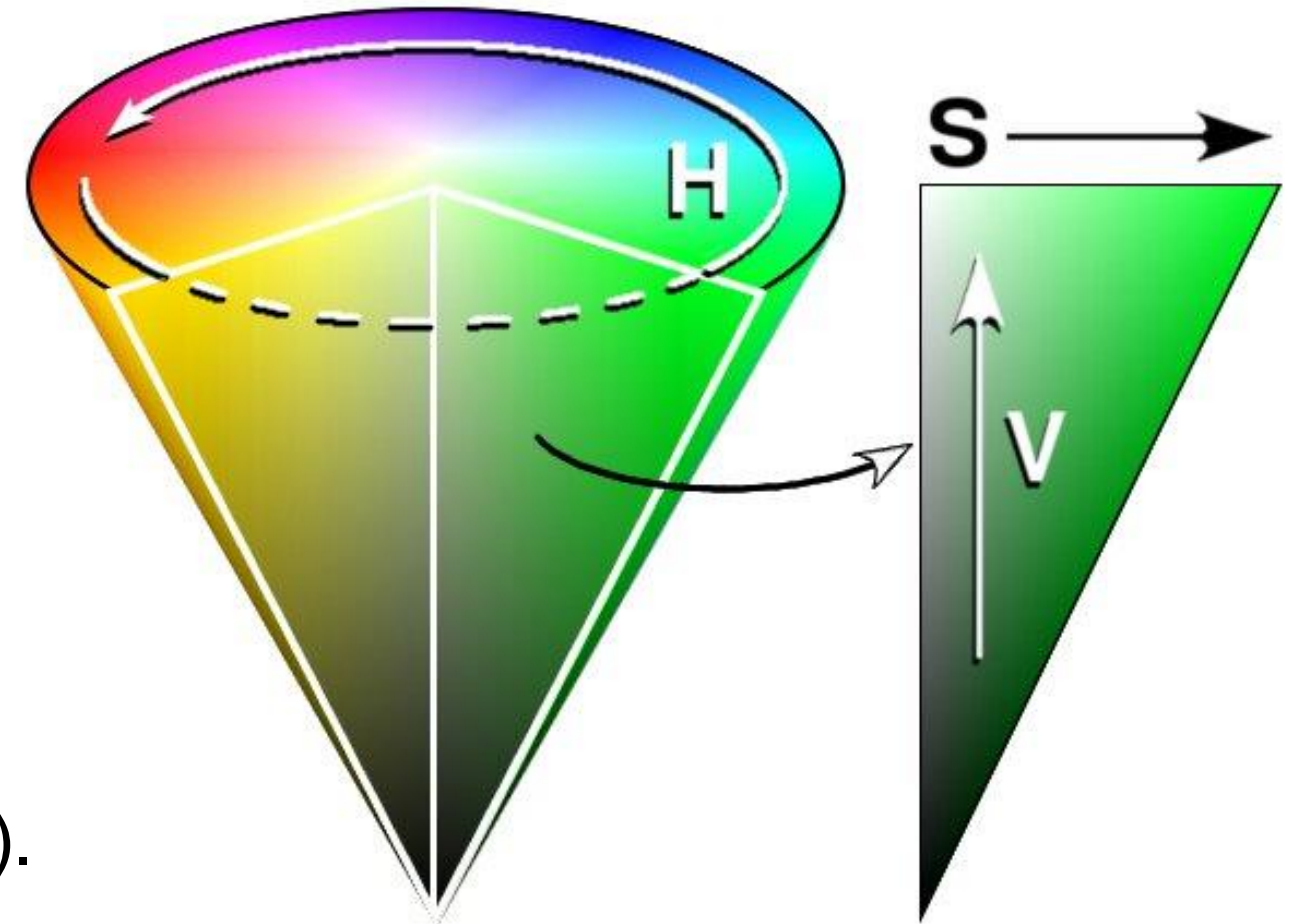
- **H (Hue)** - specifies the dominant wavelength
 - (**colour type**, e.g. red, yellow, purple ...)
- **S (Saturation)** - amount of Hue present
 - high values correspond to vibrant “**pure**” colours (when the dominant colour is mixed very little with other colours).
- **V (Value)** - brightness of colour (also known as **luminance**).



HSV Colour Space

HSV Component/Channel ranges:

- **Hue:** $0 \rightarrow 360^\circ$
 - (or $0 \rightarrow 1$ in some implementations).
- **Saturation:** $0 \rightarrow 1$
 - (e.g. for *blue*, 0.5 saturation gives sky colour and 1 gives the primary blue).
- **Value:** $0 \rightarrow 1$
 - (or $0 \rightarrow 255$ in some implementations).



RGB $\leftrightarrow$ HSV conversion is possible (*details beyond our scope*)

Example: Varying Saturation

- Bright colours (Value fixed to 1) and **decreasing values of “Saturation”** for all values of Hue.
 - *In each figure, the hue varies horizontally direction taking all possible hues.*



$s = 1$



$s = 0.75$



$s = 0.5$



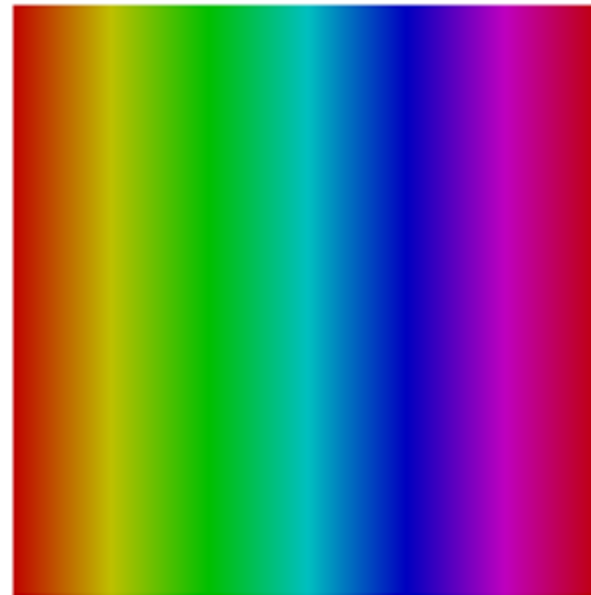
$s = 0.25$

Example: Varying Value

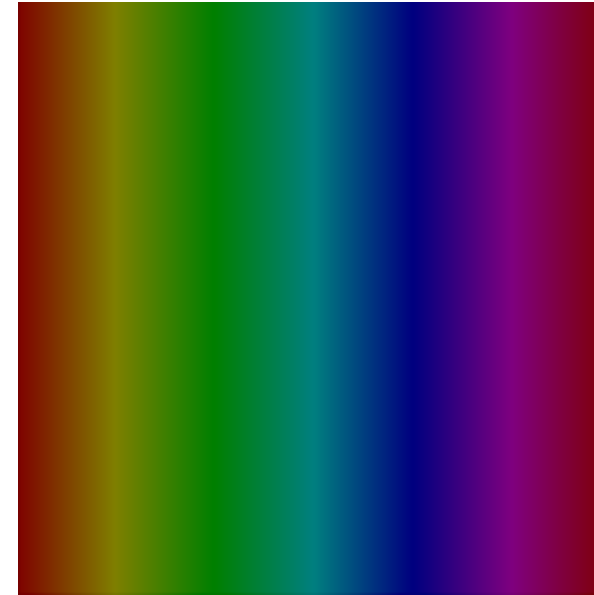
- Pure colours (Saturation fixed to 1) and **decreasing values of “Value”** for all values of Hue.
 - *In each figure, the hue varies horizontally direction taking all possible hues.*



$v = 1$



$v = 0.75$



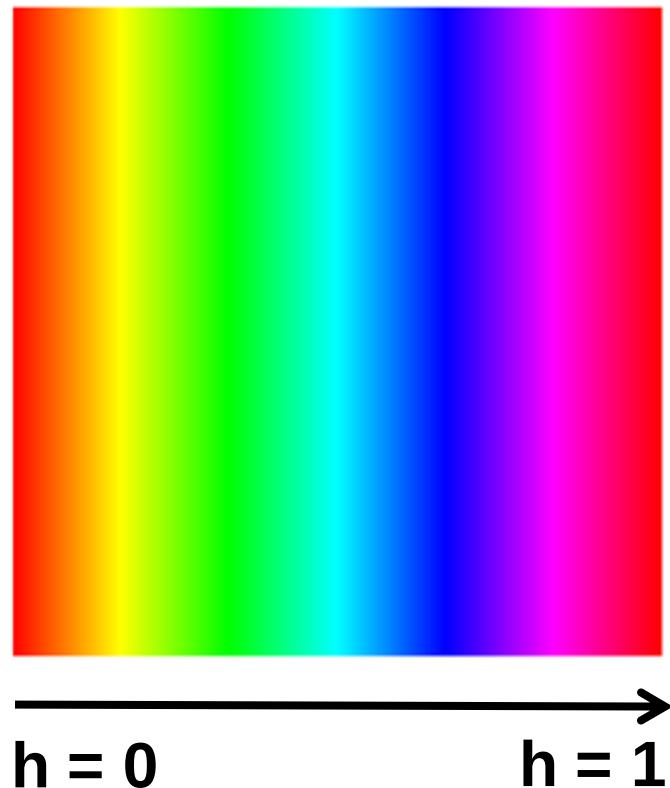
$v = 0.5$



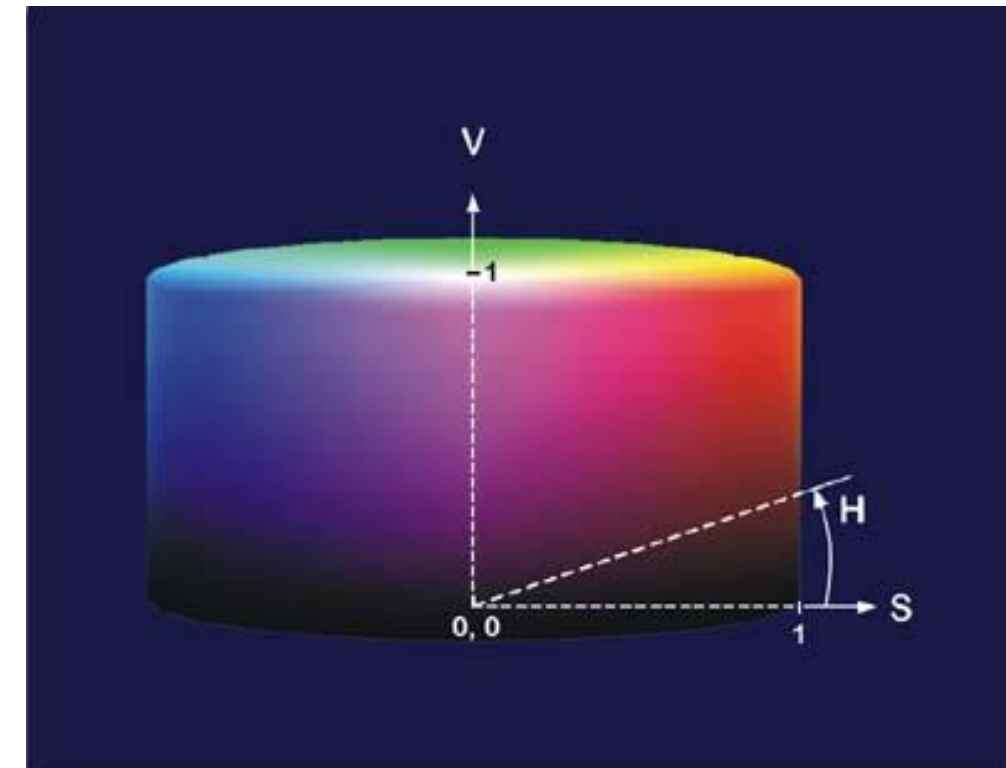
$v = 0.25$

The Geometry of the HSV Space

The hue is mapped cyclically. **Red** corresponds to $h = 0$ and $h = 360$.

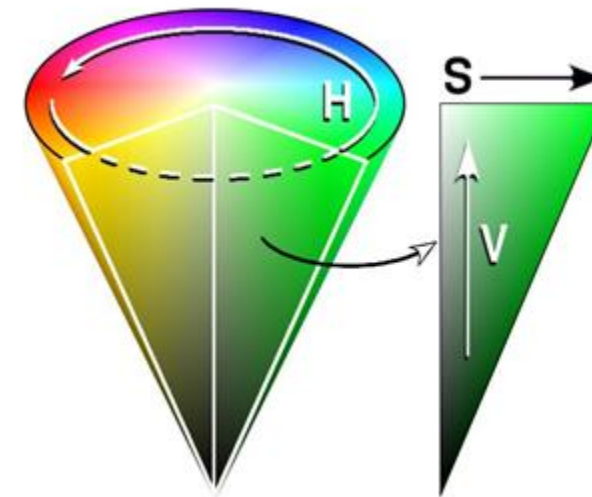
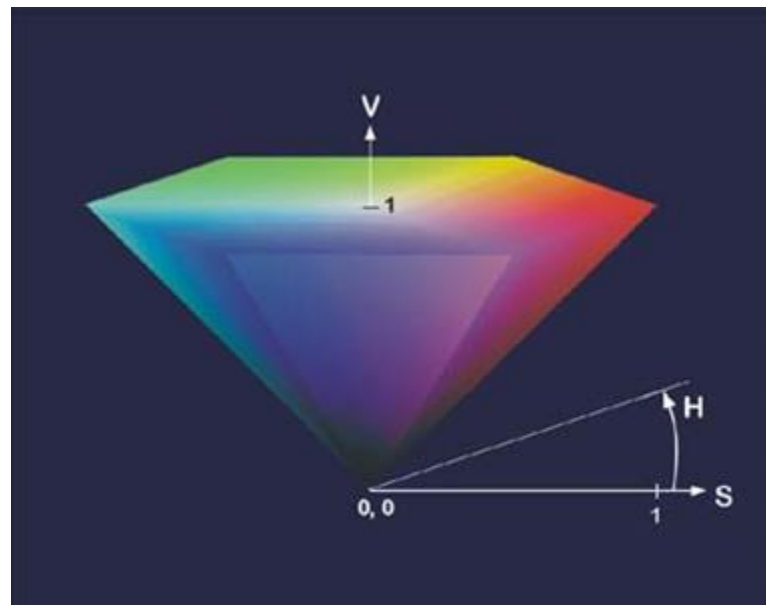


The HSV space can thus be modelled by a cylinder.

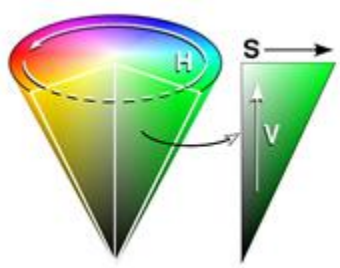


The Geometry of the HSV Space

When we **map the RGB cube onto the HSV cylinder** we get a **subset of this cylinder in the shape of a pyramid** with hexagonal base, which in turn gives rise to the cone model.



The cone model is convenient due to the fact that at low illumination levels (low Value, v) different Hues and Saturations “converge” (are difficult to differentiate).

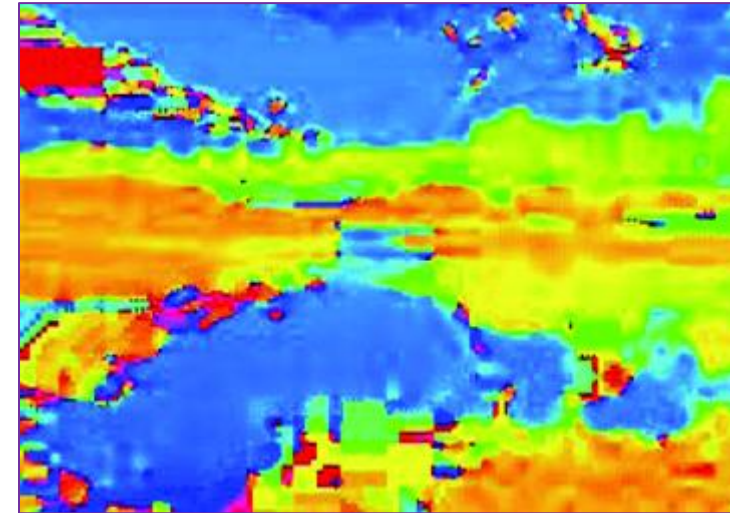


Example: HSV Channels

RGB Scene



Hue



Saturation

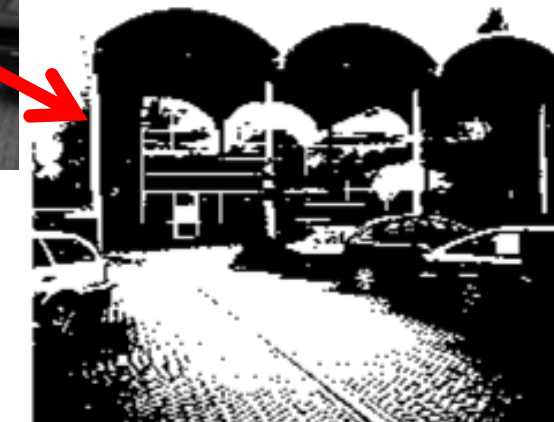


Value



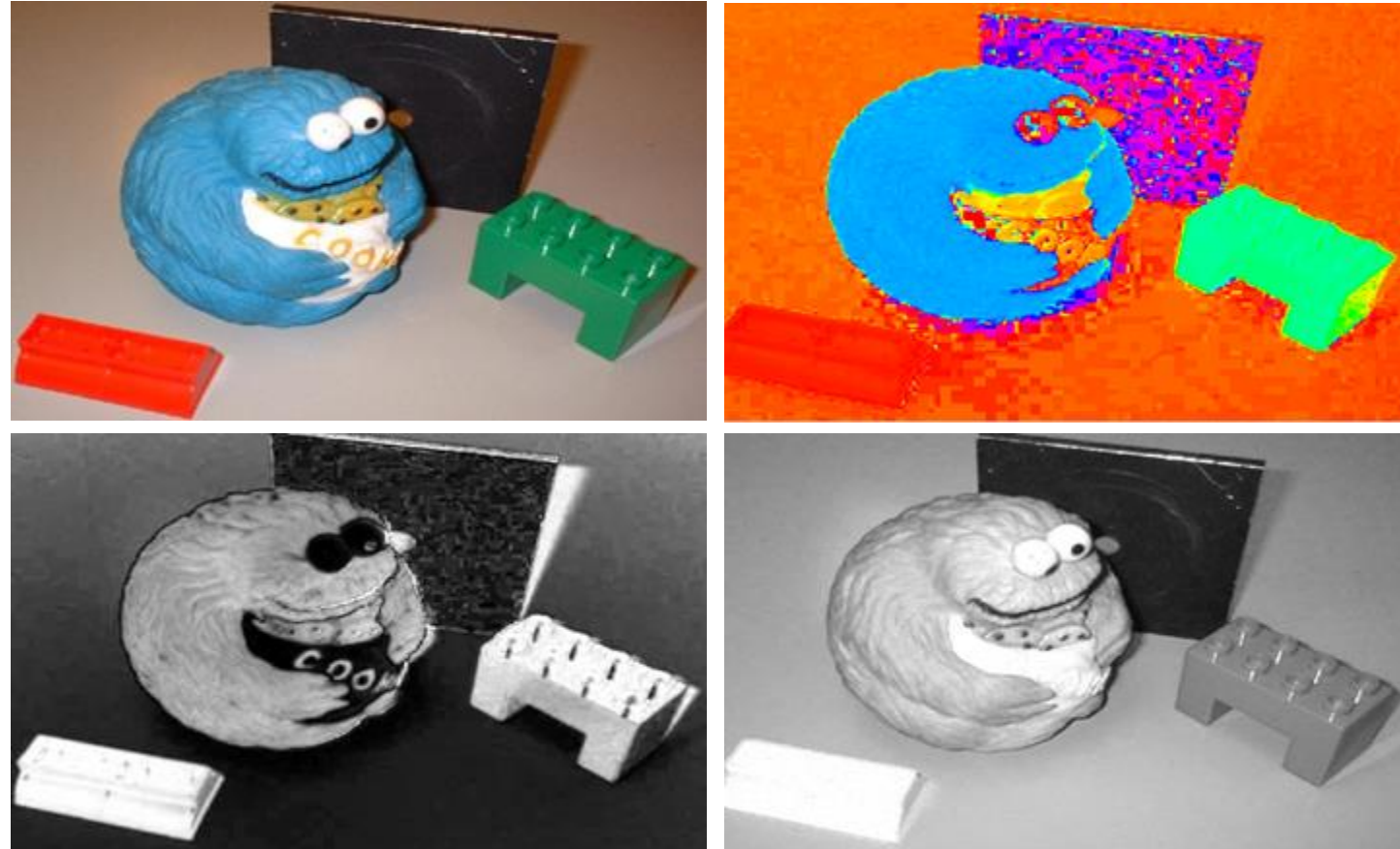
Thresholding

- **Thresholding** produces a **binary** image from a greyscale image (or colour channel).
 - Used to separate out a region or object within the image based upon its pixel values.



```
select a threshold value  $\tau$ 
for each pixel  $I(i,j)$  within the image  $I$ 
    if  $I(i,j) > \tau$ 
         $I(i,j) = 1$ 
    else
         $I(i,j) = 0$ 
    end
end
```

Thresholding in HSV

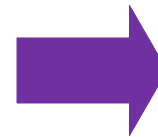


- Easier to colour threshold in HSV space than RGB.
- **Colour Slicing:** a portion of the Hue colour channel (range of wavelengths) can be isolated. Can use S and V in conjunction.

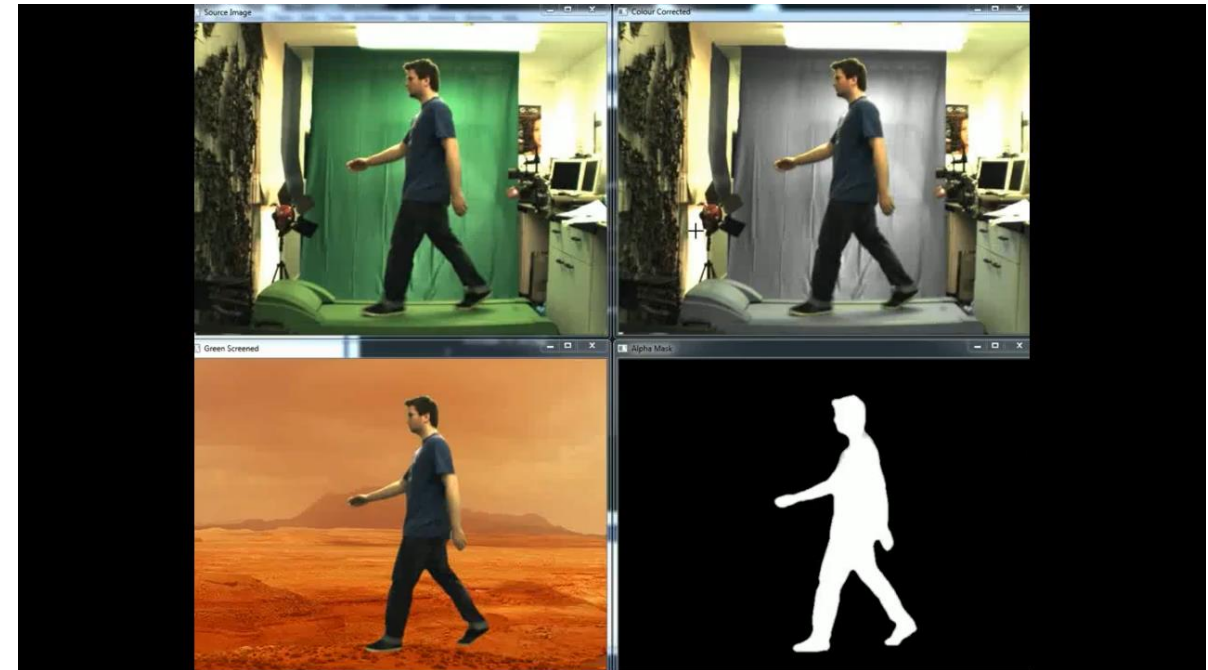
Thresholding in HSV

Isolate objects by colour:

- use an upper and a lower **threshold** on **Hue** channel to isolate by colour
- use an additional **threshold** on the **Saturation** channel to isolate pure colours
- combine the two outputs with **logical AND**.



Superman ... !



- Colour slicing (isolating colours based on Hue and Saturation) is the basis of the “blue screen” technology in movies (or “green screen”).

Cloak of Invisibility

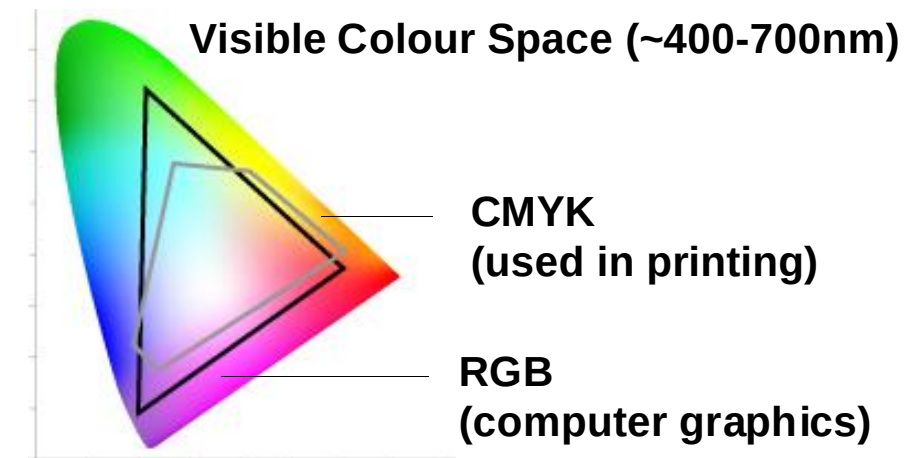
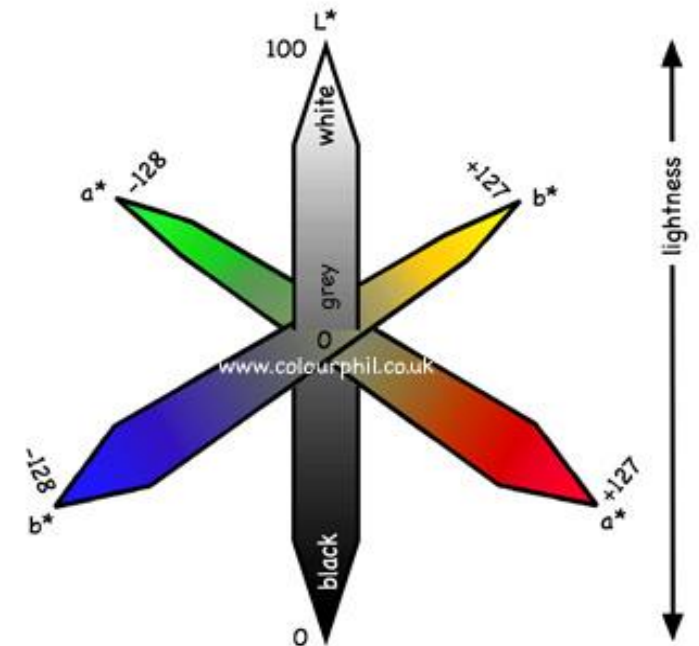


<https://github.com/atapour/invisible-me>

L*a*b Colour Space (CIELAB)

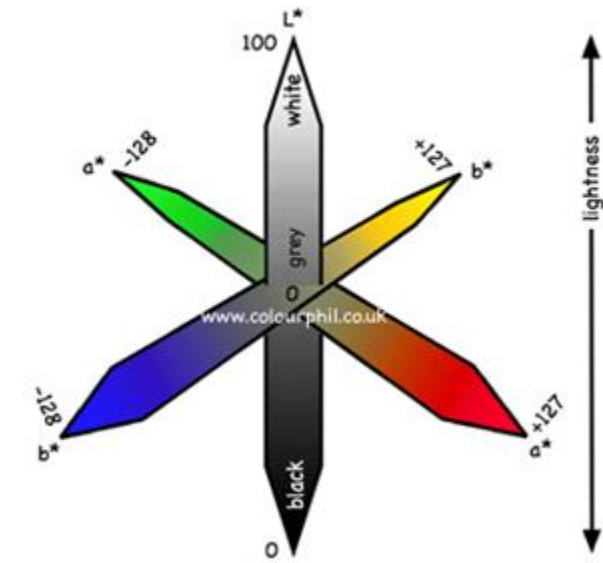
(one of the many other perceptual colour spaces that exist...)

- **Perceptually Uniform** colour space.
 - Changes in the L*a*b values are commensurate with changes in human perception of colour.
- Specified through properties of light (distributions of wavelengths).
 - Unlike RGB, colours are *device-independent*.
- Extends RGB.
 - Covers all visible colours and its parameter space can specify non-visible colours!

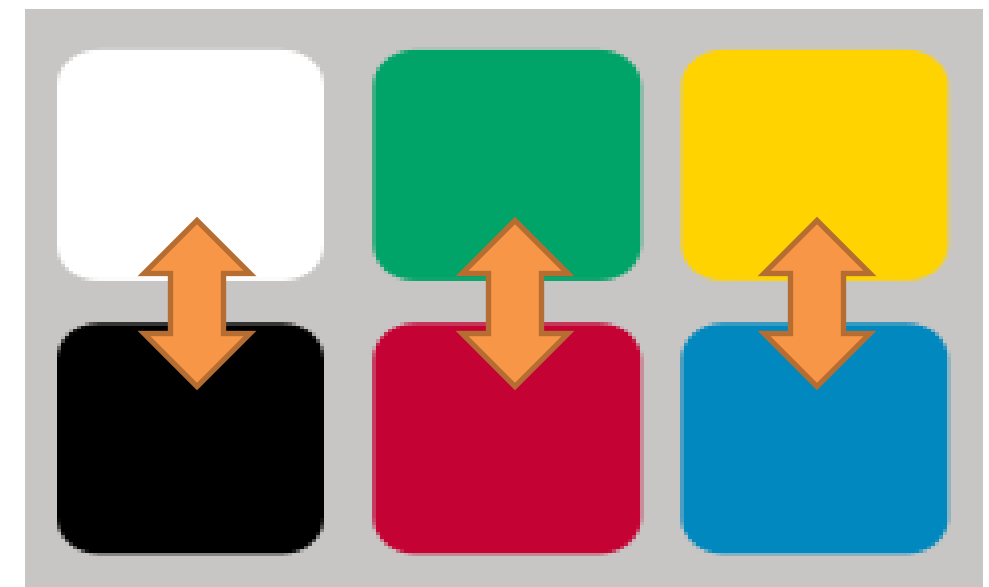


L*a*b Colour Space

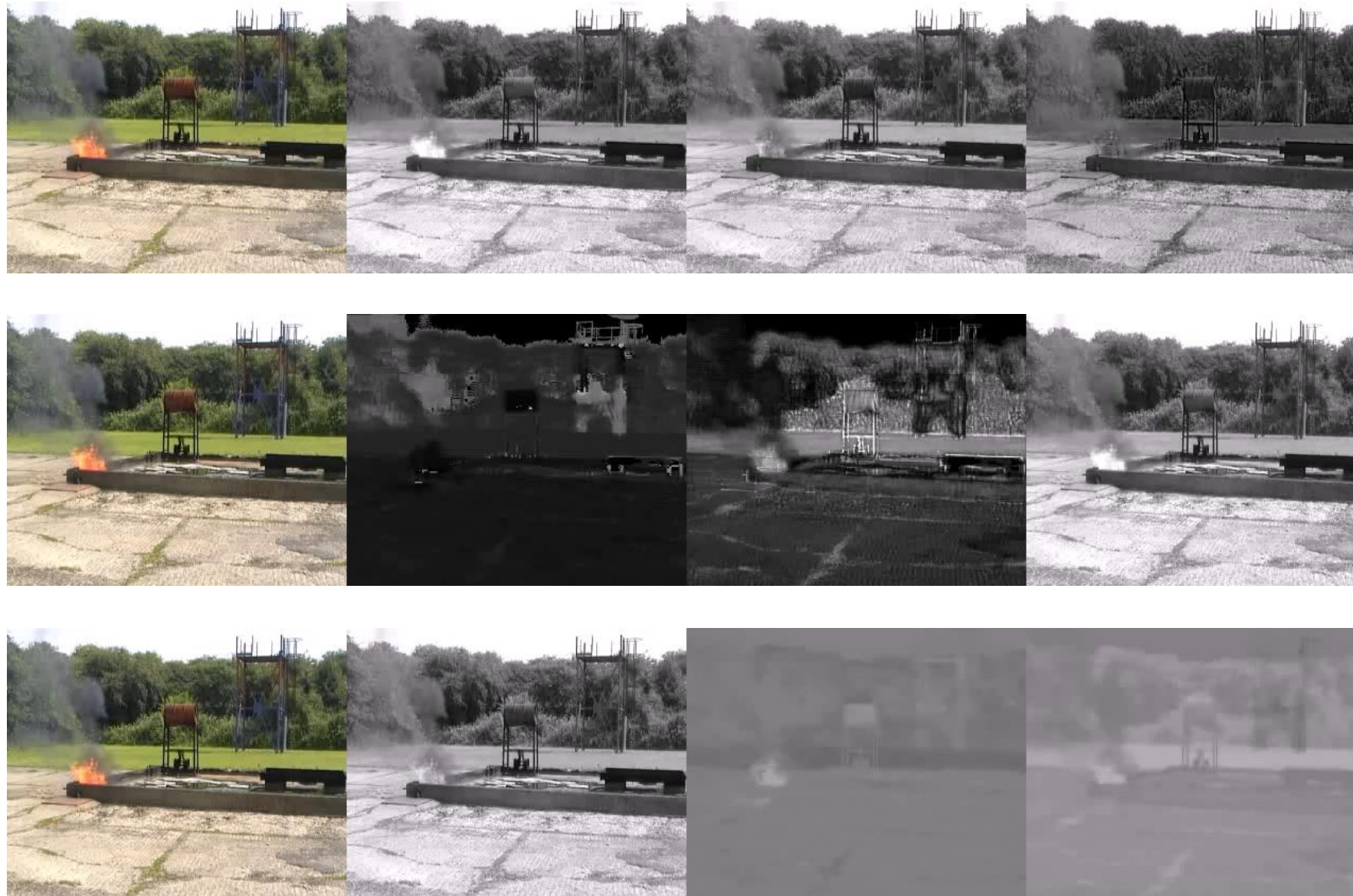
- Human vision perceives colour along **three** axes:
 - red vs. green
 - blue vs. yellow
 - black vs. white
- Why?
 - The 3 types of cone in the retina (for L, M, S wavelengths) have some **colour sensitivity overlap** thus the eye reports differences in the response of each one.
- L*a*b is often used in computer vision tasks.



L*a*b is an opponent colour space



Comparisons (RGB – HSV – L*a*b)



Quick Demonstration

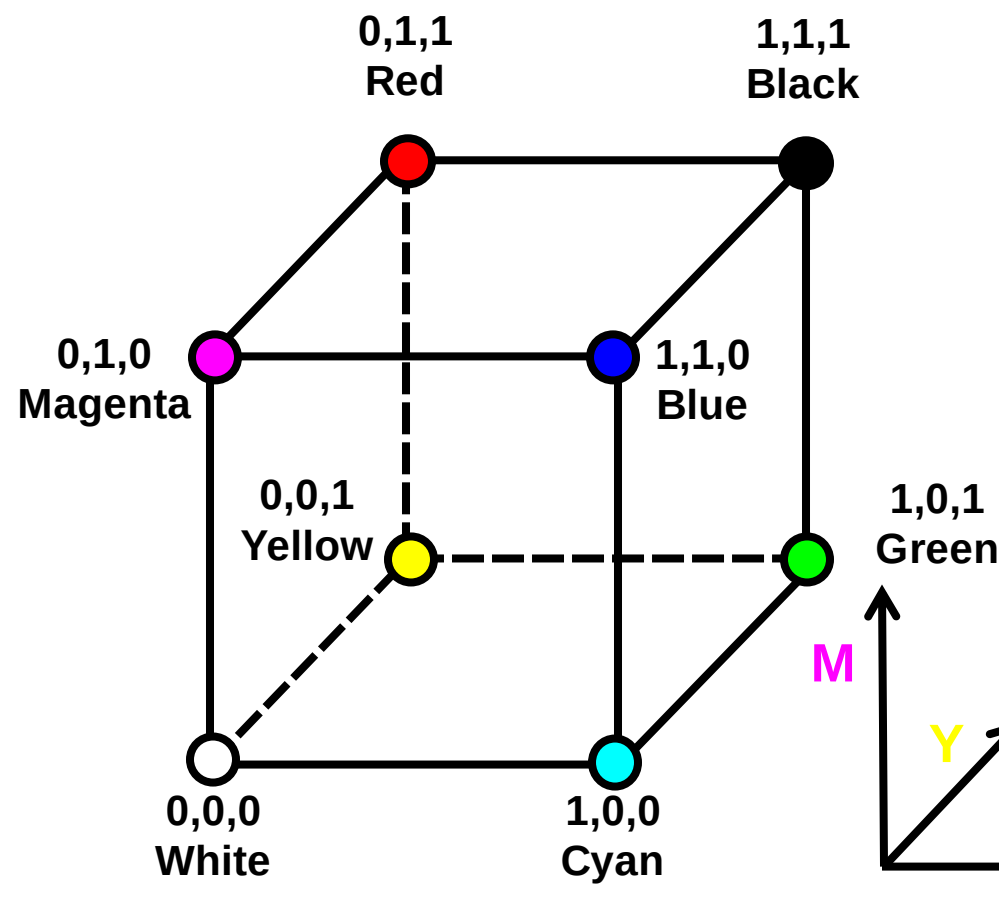
- Displaying Colour Channels of RGB, HSV and CIELAB

<https://github.com/atapour/ip-python-opencv/blob/main/colour-channels.py>



CMY Colour Space [*Printing*]

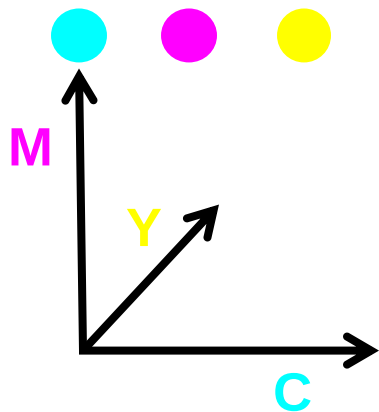
The CMY colour space is also modelled by a cube. The white is the (0,0,0) and the black is the (1,1,1).



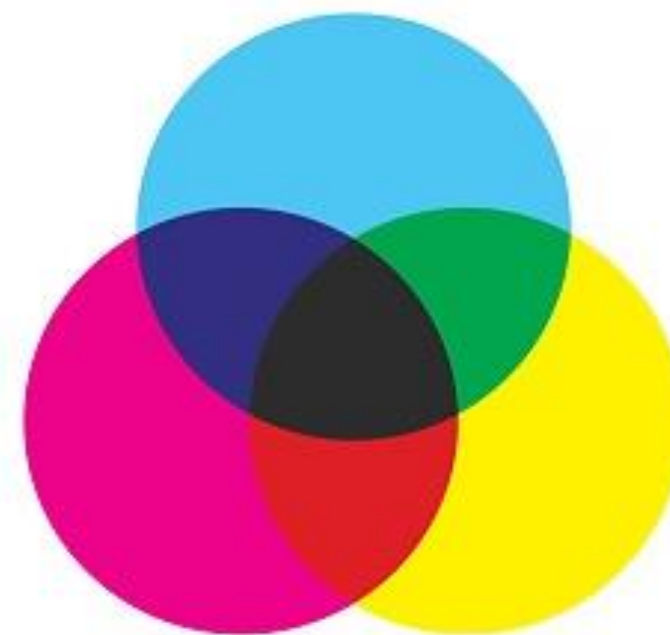
- ❑ The CMY is a **subtractive colour model** based on mixing pigments of **Cyan**, **Magenta** and **Yellow** to create other colours.
- ❑ In CMY, the colours that are seen are the part of the visible spectrum that is not absorbed by the pigments.
- ❑ e.g. yellow pigments are seen as yellow because absorb blue and reflect red and green.

CMY Colour Space [*Printing*]

- This subtractive colour space is used when applying pigments on a white background, as for example ink on paper.
 - (Usually your printer's driver does the conversion from RGB to CMY!)



Additive
(mixtures of light)
(additive primaries)



Subtractive
(mixtures of pigments)
(subtractive primaries)

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

CMYK Colour Space [*Printing*]

- ❑ The CMYK colour space is an extension of the CMY.
- ❑ Fourth component K, which is pure black and absorbs all the light in the visible spectrum.
- ❑ CMY's black (C+M+Y) is not as dark as genuine black ink: it imperfectly absorbs colour (and looks like **dark blue** or **muddy brown**...) and as a result contrast is reduced.
- ❑ Instead, four-colour **CMYK** printing gives visually superior results with greater contrast.
- ❑ CMYK also saves ink for the very common black colour!



There are many more colour spaces:
HSL, LUV, XYZ, YCbCr, YUV, ...

(the principles are largely the same!)

There is colour in colour images...

what about adding colour to greyscale images!?

Colour Mapping: False Colour

Map scalar values to colour values for display.

Simple example:

- scalar value = greyscale image value
- colour range = purple → yellow
 - ❑ Set H channel to greyscale image
 - ❑ Set V and S to fixed values

More sophisticated methods:

- intensity slicing.
- colour transfer functions.



Colour Mapping: Intensity Slicing

Consider the intensity levels

$$l_0 < l_1 < l_2 < \dots < l_M$$

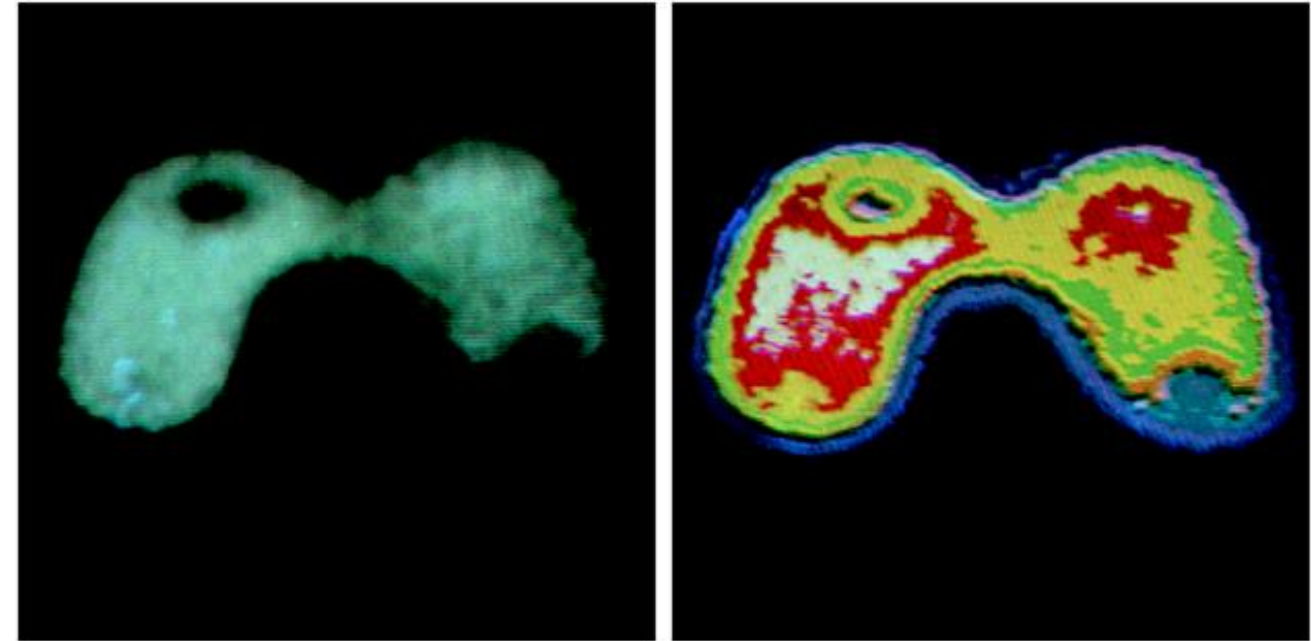
where l_0 corresponds to black and l_M to white. We use the M colours:

$$c_0, c_1, c_2, \dots, c_{M-1}$$

and a pixel with intensity s is assigned the colour c_k if

$$l_k \leq s < l_{k+1}$$

while l_M is assigned c_{M-1} .



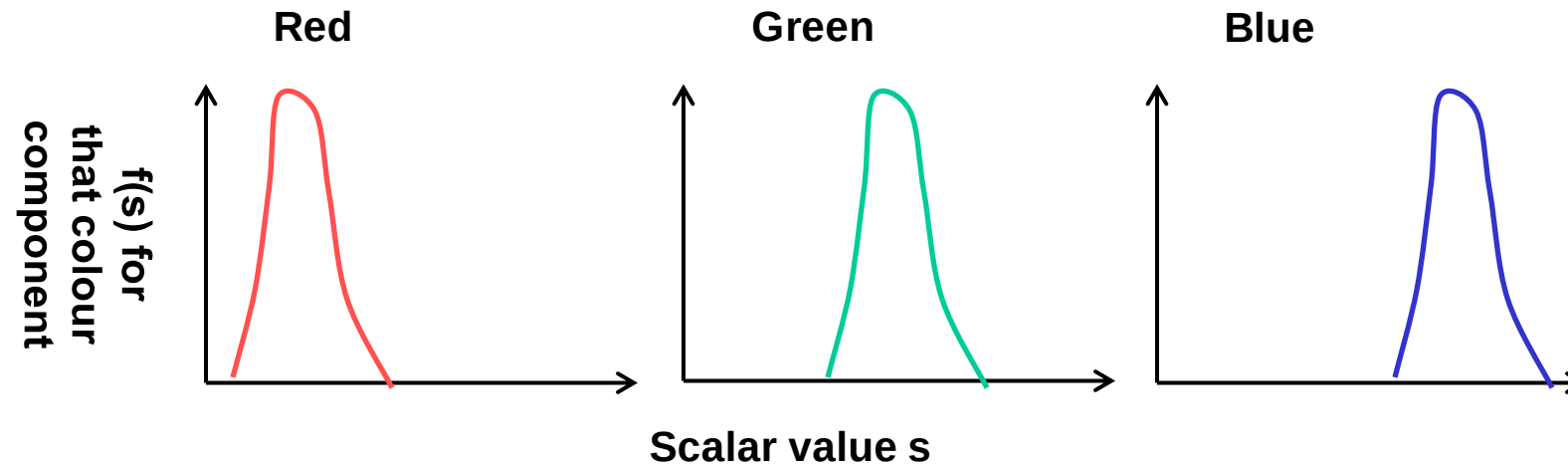
Colour Look-up Tables (LUT)

provide scalar to colour conversion
scalar values = indices into LUT
define colour transfer function

Colour Mapping: Transfer Functions

- More general technique for colour mapping:
 - scalar value s ; colour value c (usually a 3-dimensional vector)
 - colour transfer function : $f(s) = c$
- Can be used with RGB or HSV spaces.
- Any functional expression can map s into values for the colour components

c_1, c_2, c_3 :



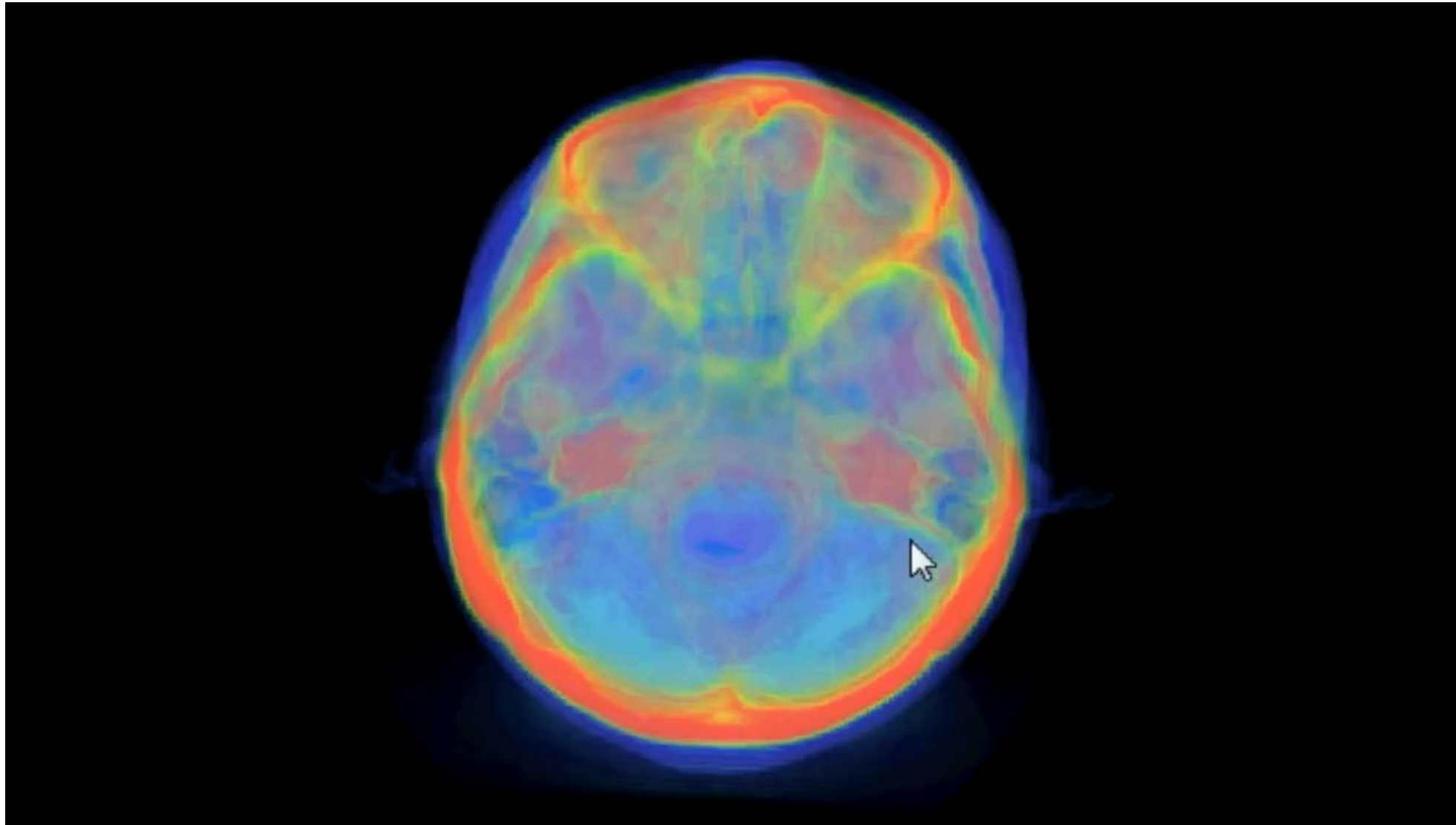
e.g. define $f()$ to convert densities to realistic skin/bone/tissue colours in 3D medical imaging

Q: Why add false colour ?

How many greys can you see? **30 - 900**

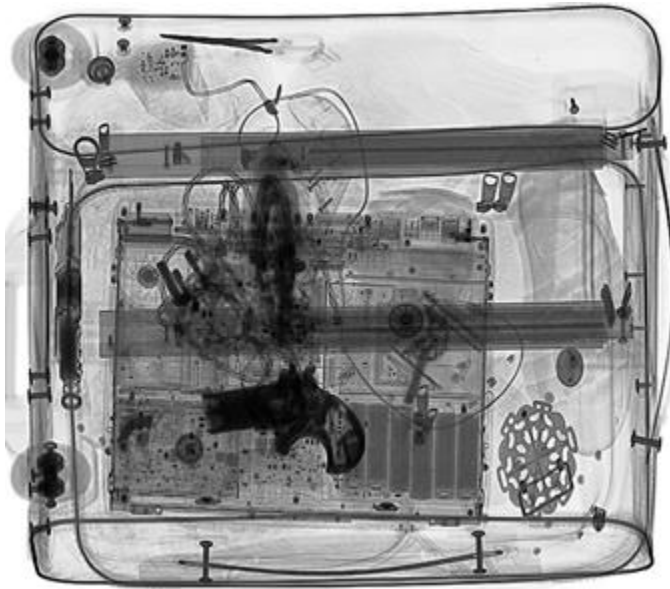
How many colours can you see? **1 – 10
million**

Application - Medical Imaging

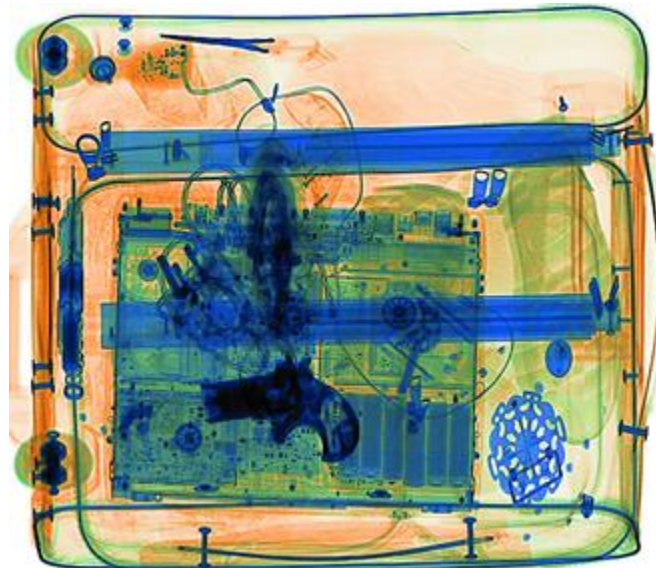


False colour mapping of CT medical image scan

X-Ray Security Screening



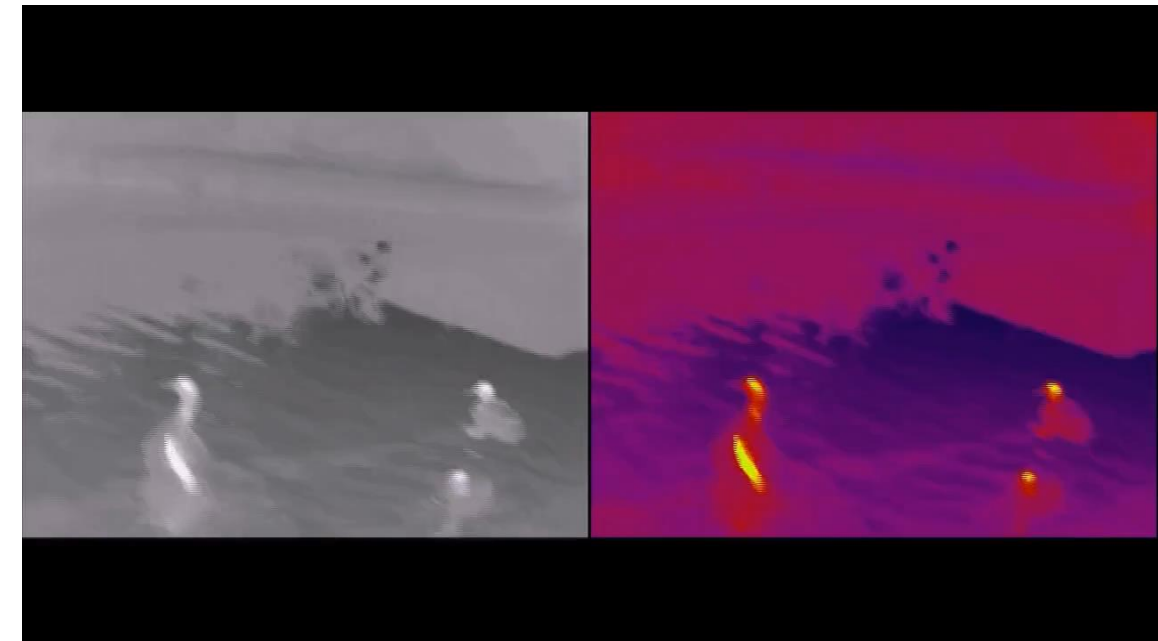
- X-ray image of a bag from an airport scanning system.
- Raw X-ray is greyscale (one value per pixel returned from the scanner).
- Add false colour to allow different materials to be identified and improve screening performance.



Source: [Turscany, 2013]

Infra-Red Thermal Imagery

- Infra-red intensity received from sensor as 2D array of infra-red readings.
 - Visualised using colour mapping (mapped to Hue) calibrated to temperature.

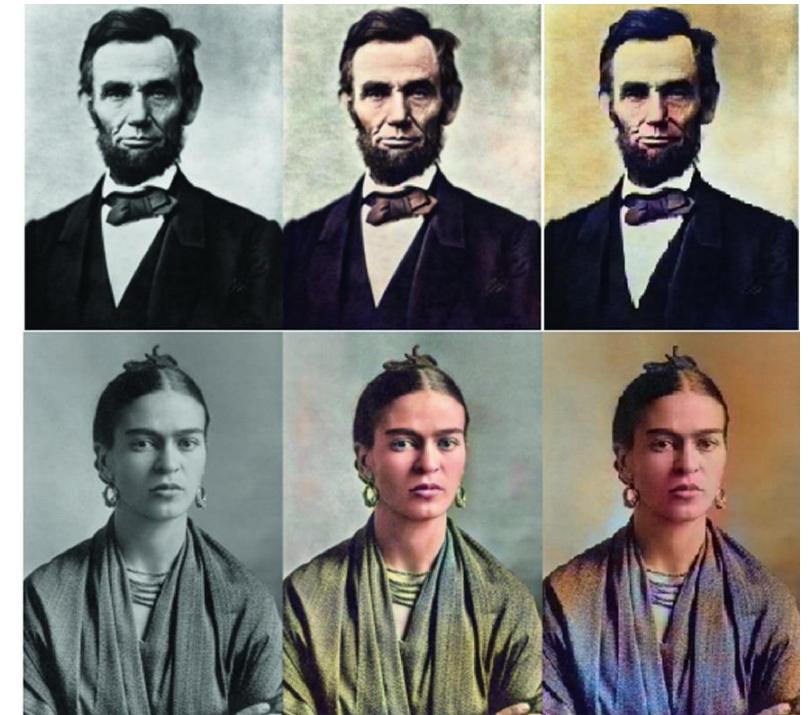


Colourising Grayscale Images

- Adding colours to grayscale photos artificially – normally involves human intervention or a learning-based model.
 - The CIELAB colour space is normally used.



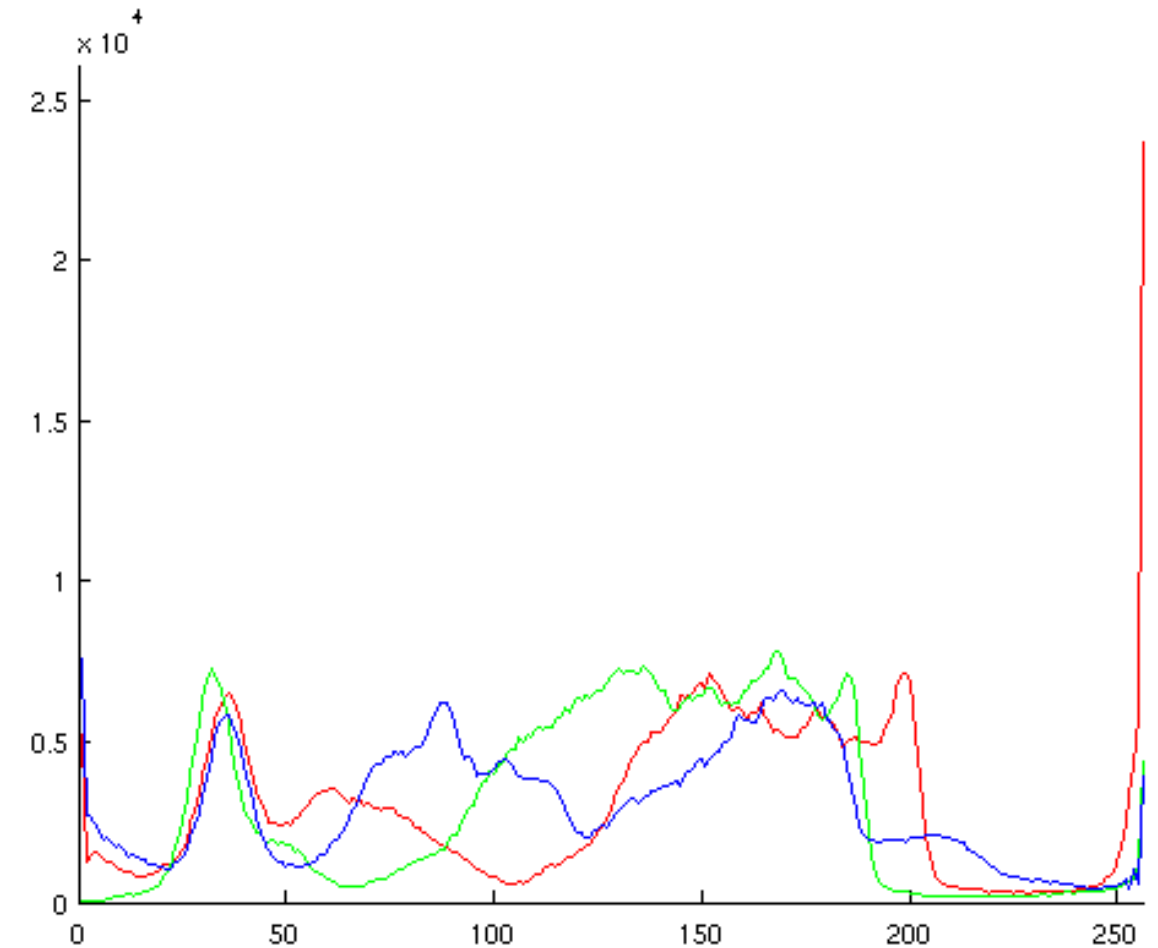
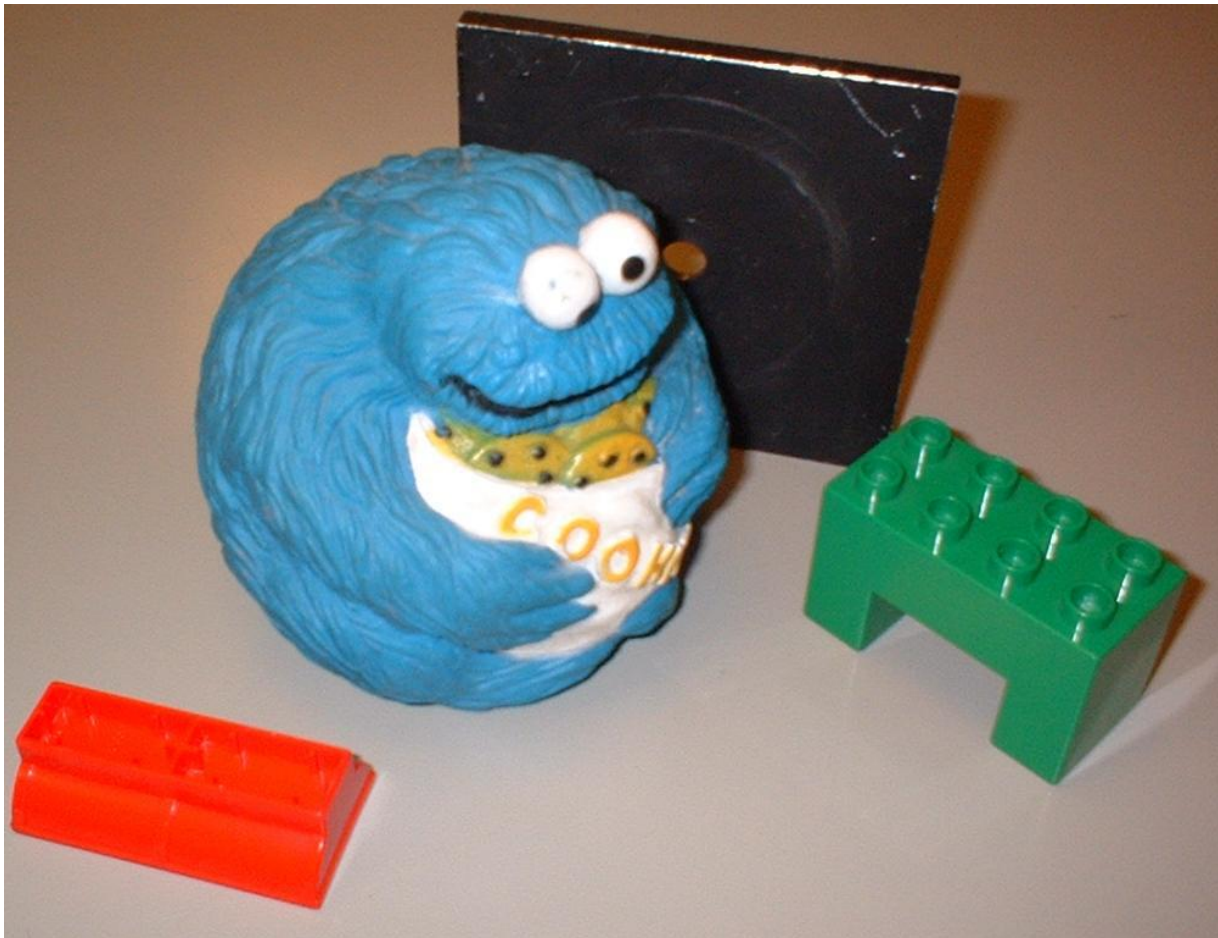
[Zhang et al., 2017]



[Abualola et al., 2021]

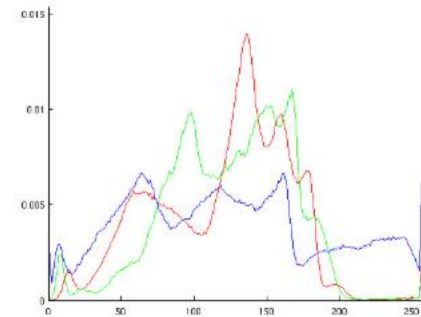
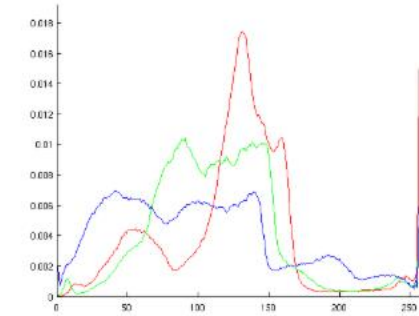
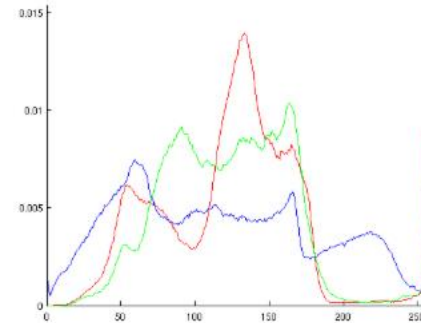
Recap - RGB Colour Histograms

- Histogram of each R, G, B colour channel (separately).

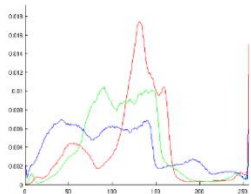


Histograms: Statistical Distributions

- Histogram = statistical distribution of image content
 - no structural representation
 - **different images** have very **different histograms**
 - **similar images** have **similar histograms**



- Histogram Comparison
 - compare stat. dist. of images
 - use normalised histograms
 - e.g. chi-squared, Euclidean ...



Histograms Similarity Distances

(between image histograms H_1 and H_2)

■ Correlation:

$$d(H_1, H_2) = \frac{\sum_{i=1}^N H'_1(i) - H'_2(i)}{\sqrt{\sum_{i=1}^N [H'_1(i)]^2 \sum_{i=1}^N [H'_2(i)]^2}}$$

$$H'_k = \frac{H_k(i) - 1}{N \sum_{i=1}^N H_k(i)}$$

~1 means image similarity

■ Chi-squared:

$$d(H_1, H_2) = \sum_{i=1}^N \frac{H_1(i) - H_2(i)}{H_1(i) + H_2(i)}$$

~0 means image similarity

■ Intersection

$$d(H_1, H_2) = \sum_{i=1}^N \min(H_1(i), H_2(i))$$

~1 means image similarity

■ Bhattacharyya

$$d(H_1, H_2) = \sqrt{1 - \sqrt{\sum_{i=1}^N H_1(i) - H_2(i)}}$$

~0 means image similarity

Histograms Similarity Distances

Application: basic image search

Image 0



Image 1



Image 2



Query Image

image 0 Corr: 0.187 ChiSq: 1.047 Intersect: 0.327 Bhatt: 0.596
 image 1 Corr: 0.713 ChiSq: 0.398 Intersect: 0.606 Bhatt: 0.346
 image 2 Corr: **0.974** ChiSq: **0.065** Intersect: **0.889** Bhatt: **0.135**

Histograms Similarity Distances

Application: basic image search

Image 0



Image 1



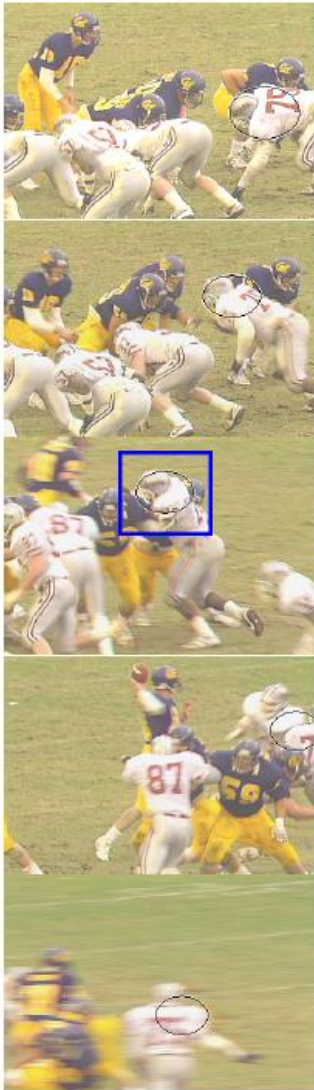
Image 2



Query Image

image 0 Corr: 0.189 ChiSq: 0.857 Intersect: 0.407 Bhatt: 0.525
 image 1 Corr: 0.814 ChiSq: 0.181 Intersect: 0.740 Bhatt: 0.223
 image 2 Corr: **0.931** ChiSq: **0.184** Intersect: **0.793** Bhatt: **0.266**

Example: Basic Object Tracking



- extract histogram of object / region of interest
- search for histogram patch within image (using statistical comparison)
- use robust statistical method (mean shift) to isolate / track region
 - e.g. using **Hue and Saturation** from HSV
- Key advantage: non-rigid objects
- Key disadvantage: colour variance of object

Quick Demonstration

- Object Tracking – Mean Shift based on Hue and Saturation from HSV:

<https://github.com/atapour/ip-python-opencv/blob/main/colour-object-tracking.py>



Colour Image Processing



RGB Colour Image →

Logical
Arithmetic
(lecture 2)

Spatial
Filtering
(lectures 4 + 5)

Fourier Filtering
and Correlation
(lectures 7 + 8)

Apply to individual
colour channels in
any colour space
representation.

may convert to
greyscale to reduce
computation if
colour output is
not required.



Colour Image Processing



RGB Colour Image →

Contrast
Enhancement
(lecture 3)

Histogram Based
Transforms
(lecture 6)

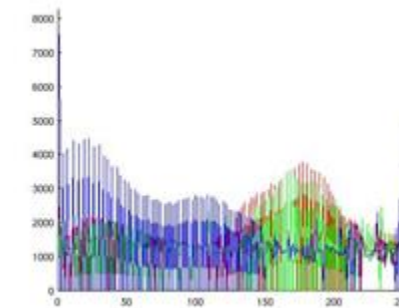
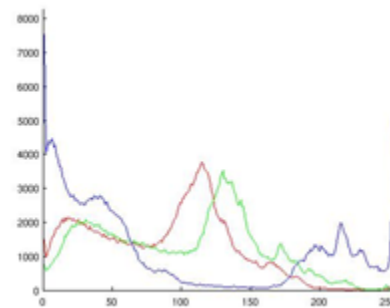
+ this lecture

For processes like **altering the dynamic range** of the image, in order to preserve the relative weights of the colour information in the image, we first convert to a perceptual colour space that isolates the scene illumination in a single channel (e.g. V of HSV or L of L*a*b).



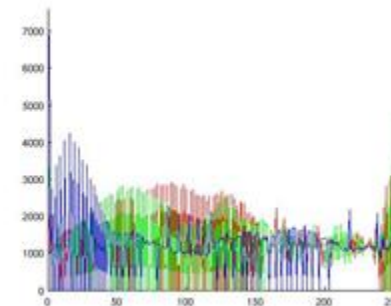
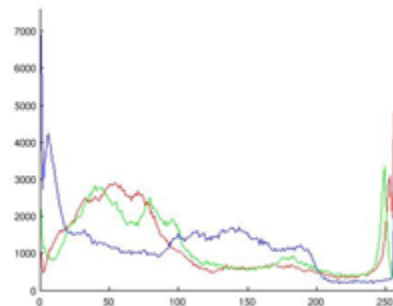
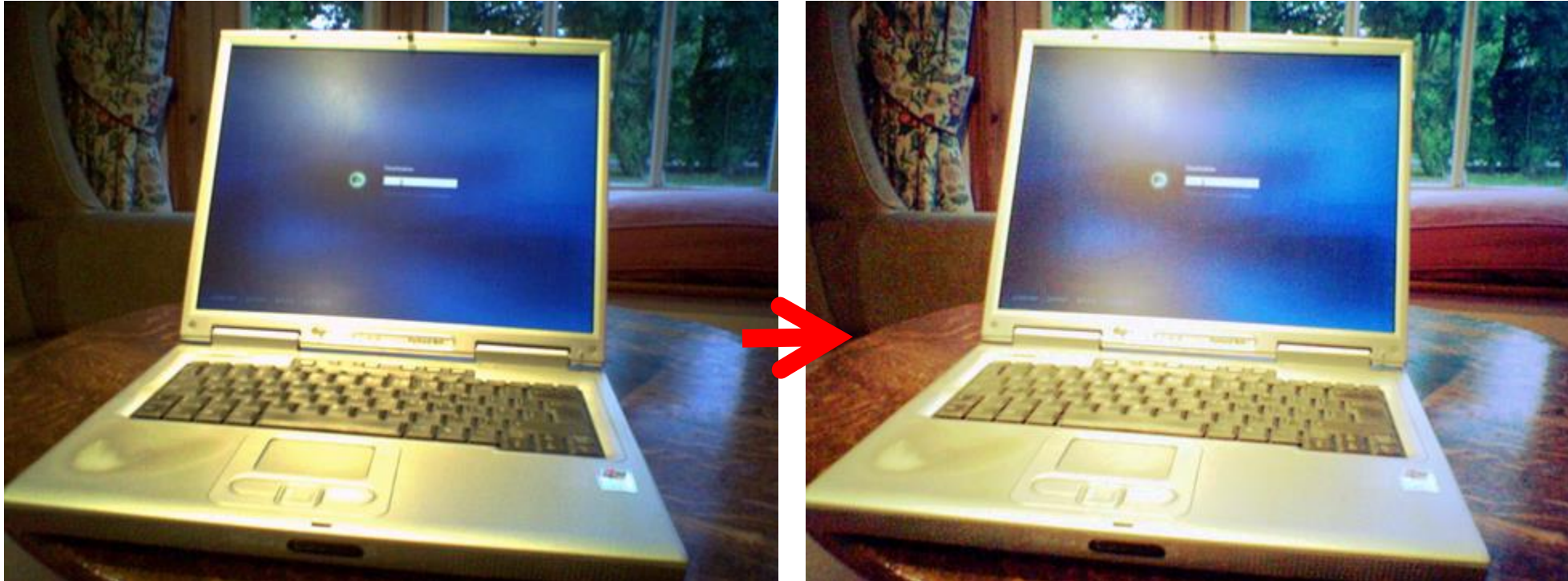
We subsequently perform enhancement on this channel in isolation, add it back to the other channels that contain the image colour (chromatic) information and transform back to RGB.

Example: Histogram Equalisation



Histogram equalisation on each RGB channel gives strange results

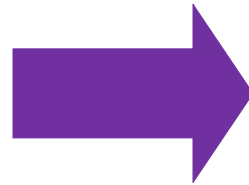
Example: Histogram Equalisation



Histogram equalisation on each RGB channel gives strange results

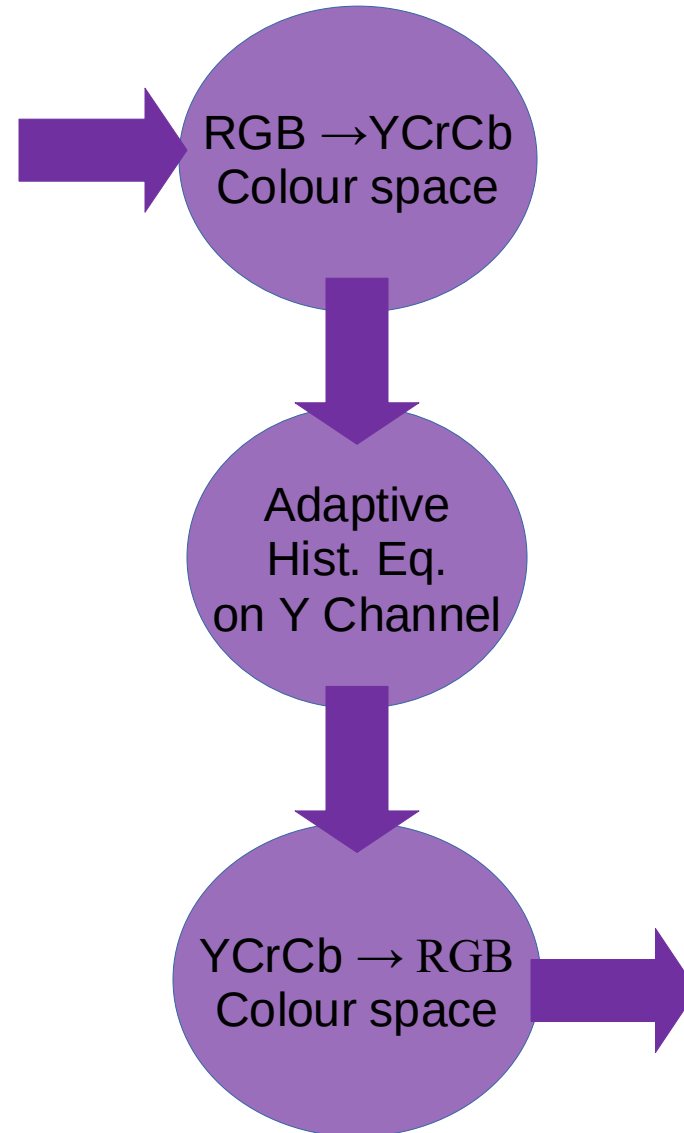
Example: Histogram Equalisation

1. RGB $\rightarrow$ HSV
2. Histogram Equalize V channel
3. HSV $\rightarrow$ RGB



Histogram equalisation on V channel of HSV
*Similarly on L (of L^*a^*b) or Y (of YCrCb)*

Example: Histogram Equalisation

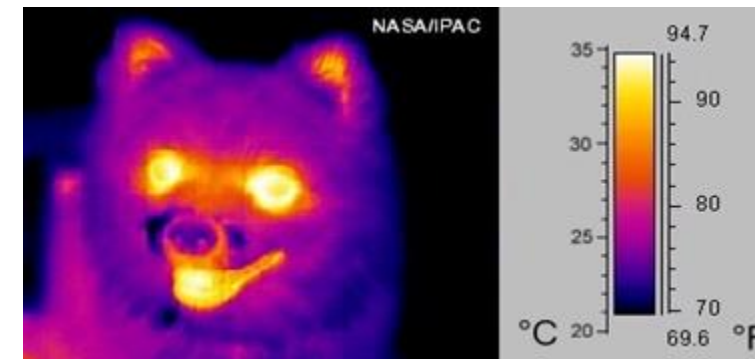
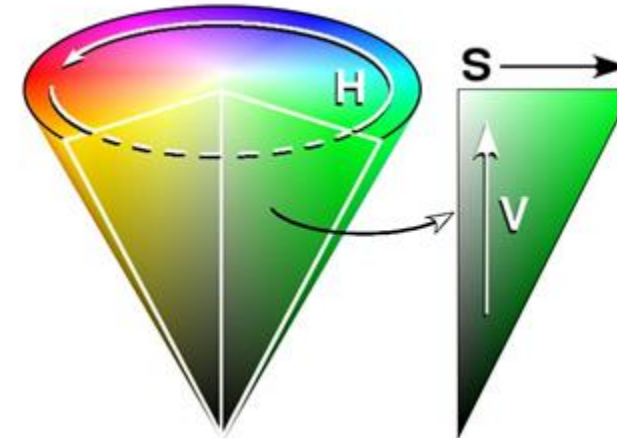


Source: J. Brown, Cranfield Health / Gloucestershire NHS

Summary

Colour spaces

- RGB
- HSV
- L^*a^*b
- CMY
- CMYK



False colour mapping

Colour image processing

